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Studierichting: Informatica
Oefeningenreeks 6E nr. 11.3

$f(x) = x^8 + 8x^7 + 12x^6 - 40x^5 - 74x^4 + 120x^3 + 108x^2 - 216x + 81$

$f'(x) = 8x^7 + 56x^6 + 72x^5 - 200x^4 - 296x^3 + 360x^2 + 216x - 216$

$\Leftrightarrow f'(x) = 8(x-1)^3(x+1)(x+3)^3$

$f'(x) = 0 \Leftrightarrow x = 1$ of $x = -1$ of $x = -3$

onderzoek in $x = 1$:

$f'(1) = f''(1) = f'''(1) = 0$

$f^{(4)}(1) = 6144 > 0$

$\Rightarrow$ lokaal minimum in $(1, 0)$

onderzoek in $x = -1$:

$f'(-1) = 0$

$f''(-1) = -512 < 0$

$\Rightarrow$ lokaal maximum in $(-1, 256)$

onderzoek in $x = -3$:

$f'(-3) = f''(-3) = f'''(-3) = 0$

$f^{(4)}(-3) = 6144 > 0$

$\Rightarrow$ lokaal minimum in $(-3, 0)$

References